An Educational Framework for AI-Driven RF Signal Processing on FPGA

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***Abstract*—University electrical engineering programs typically deliver instruction in FPGA digital design, RF communications, embedded systems, and machine learning as independent courses, leaving graduates without structured experience integrating these disciplines. This paper presents an educational framework built around a benchtop hardware platform in which two STM32+RFM95W cooperative target stations exchange DATA and ACK traffic on a shared 906.5 MHz channel in the U.S. 915 MHz ISM band, while a third STM32+RFM95W interceptor station passively captures the bidirectional traffic and forwards it to a Digilent Nexys A7-100T FPGA, with collected signal data feeding a machine learning classification pipeline. The interceptor additionally operates a button-gated PAUSE protocol that selects the highest-RSSI target and commands a temporary firmware-level transmit hold-off, providing a closed-loop intercept-and-act behavior without requiring transmissions outside FCC ISM-band rules. A five-module progressive curriculum sequences students from hardware bring-up and embedded firmware development through FPGA RTL design, system integration, AI-driven signal analysis, and full-system stress testing, benchmarking, and failure-mode analysis. Each module makes the interfaces between disciplines explicit teaching objectives, enabling students to reason about how design decisions in one domain propagate constraints into adjacent ones. The validation module culminates the framework by deriving the system specification empirically—through structured stress tests, six-point latency benchmarking, closed-loop PAUSE behavioral measurements, and a cross-domain failure-mode catalog—rather than asserting it a priori.**

***Index Terms*—FPGA, LoRa, machine learning, RF signal processing, embedded systems, curriculum design, signal classification**

I. Introduction

Modern electrical and computer engineering practice increasingly demands fluency across multiple technical domains simultaneously. Defense, telecommunications, and aerospace systems routinely integrate real-time RF data acquisition, hardware-accelerated digital processing, embedded firmware, and intelligent signal analysis within a single pipeline. University curricula, however, frequently deliver these disciplines in isolation. A student may learn HDL synthesis and timing closure in a digital design course, modulation theory and channel models in a communications course, interrupt-driven firmware in an embedded systems course, and classification on benchmark datasets in a machine learning course—without encountering a structured experience that requires synthesizing these skills into a working system.

This isolation creates a gap between academic preparation and professional practice. Engineers who understand each domain individually often struggle with the interface problems that arise when combining them: how RF modulation parameters affect FPGA buffer sizing, how reconfigurable logic resource constraints shape a data collection pipeline for machine learning, or how real-world RF propagation deviates from classroom channel models. These cross-disciplinary concerns are not incidental details but core engineering competencies that no single-domain course addresses.

Prior research has contributed to individual components of this problem but has not combined them into an integrated educational experience. On the pedagogical side, cloud-based FPGA toolchains with structured lab sequences have lowered the barrier to digital design education [9], and the STM32 Nucleo platform has been validated as an effective embedded systems teaching tool with significantly greater peripheral capability than Arduino-class boards [10]. On the technical side, FPGA-based RF signal processing and inference have advanced considerably: Boolean reservoir computing achieves CNN-competitive modulation classification accuracy with far fewer trainable parameters [1], streaming CNN architectures on RFSoC FPGAs classify eight modulation schemes continuously at 128 MHz [3], FPGA-accelerated ResNet reduced angle-of-arrival inference cycle count by 28.2% over CPU while improving robustness to multipath fading [2], and SoC FPGA LoRa gateways have demonstrated hardware-accelerated processing within tight resource budgets [7]. For LoRa signal classification specifically, deep learning has proven effective across multiple tasks: CNNs on spectrogram inputs achieved 99.77% spreading factor detection across a -20 dB to +30 dB SNR range [5], CNNs on raw I/Q data reached 93.3% device identification accuracy under real experimental conditions [6], neural network demodulation matched coherent matched-filter performance while tolerating carrier frequency offset [8], and comparative studies of DNN, CNN, and DBN architectures provide empirical guidance on accuracy-complexity tradeoffs for resource-constrained deployment [4]. Each of these works advances the state of the art within its own discipline by reporting peak accuracy,

throughput, or resource utilization on a curated, single-domain task. None of them is—or attempts to be—a curriculum that develops the cross-domain reasoning skills required to build, deploy, and reason about an integrated system. What is absent is a curriculum that sequences these subsystems into a unified learning experience where the connections between them are the primary subject of instruction, and that produces the categorically different evidence (cross-module regression survival, integration-failure root-cause depth, closed-loop intercept-and-act behavior) that single-domain accuracy numbers architecturally cannot.

The contributions of this paper are: (1) a five-module progressive educational curriculum that integrates FPGA digital design, RF communications, embedded firmware, and machine learning into a single hardware-grounded learning sequence; (2) a concrete benchtop reference platform built from commercially available components that instantiates each cross-domain interface as an experimentally observable design problem, including a button-gated PAUSE protocol that exercises closed-loop intercept-and-act behavior between the interceptor and the cooperative targets without violating FCC ISM-band rules; and (3) a comparison framework for evaluating the platform's outcomes against single-domain prior work, in which the framework's distinctive value is shown through measurements (chain-depth distribution, regression survival, closed-loop response latency) that single-domain studies are architecturally unable to report. Section II formally defines the research problem and its context. Section III presents the framework in detail. Section IV presents the expected and structured-measurement results across all five modules. Section V situates the framework's outcomes against the single-domain prior work cited above.

II. Problem Statement

The research problem is the absence of a structured, progressive educational curriculum that integrates FPGA digital design, RF communications, embedded microcontroller programming, and machine learning-based signal analysis into a unified learning experience where each cross-domain interface is a deliberate teaching objective.

The problem has a strict sequencing constraint that single-domain curricula do not impose. A student cannot meaningfully study the RF-to-FPGA interface without first having a working bidirectional cooperative-target link, cannot study the FPGA-to-data-pipeline interface without first having a working RF-to-FPGA interface, and cannot train a useful classifier without first having a labeled dataset collected from that pipeline. Hardware bring-up must therefore come first, and every downstream learning objective inherits the correctness of what preceded it. This dependency chain is the reason the framework is organized as a progressive module sequence rather than a set of parallel tracks.

The target population is senior undergraduate and master's-level electrical engineering students who have completed introductory coursework in each of the four domains individually. These students possess the foundational prerequisites—HDL syntax, basic modulation and link theory, embedded C programming, and introductory machine learning—but lack experience working with systems that require all four simultaneously. When confronted with a multidisciplinary design problem, they face a secondary difficulty: understanding how a decision made within one domain generates requirements and constraints in adjacent domains that cannot be resolved from within a single discipline.

The concrete system context is a benchtop RF signal processing testbed. Two STM32 Nucleo-L476RG microcontrollers, each paired with an RFM95W LoRa transceiver, operate as cooperative target stations (Target A, ID `0x01`; Target B, ID `0x02`) that exchange DATA and ACK packets on a shared 906.5 MHz channel: Target A initiates a DATA transmission on a configurable schedule, and Target B responds with an ACK whose sequence number echoes the received DATA. A third STM32+RFM95W node operates as the interceptor station (Threat, ID `0xFF`) and passively captures every DATA and ACK frame it can demodulate on the same channel, forwarding parsed metadata to a Digilent Nexys A7-100T FPGA. The FPGA performs real-time detection, packet integrity verification, and feature extraction in programmable logic on the Artix-7 fabric. Extracted signal features are collected by a host machine and used to train and evaluate machine learning classifiers for signal classification, packet-type identification, and emitter identification tasks. The interceptor also implements a button-gated PAUSE protocol: while a user button on the interceptor board is held, the interceptor transmits a PAUSE packet at a 100 ms cadence addressed to the highest-RSSI target currently in its observation table, and the targeted station's firmware enters a 500 ms transmit hold-off and prints `"I'm Jammed"` on its debug serial port. The PAUSE behavior is implemented entirely as a firmware-level convention over standard ISM-band LoRa transmissions and uses no out-of-band emissions or jamming techniques that would require special licensing. This architecture exposes four concrete interface problems that span disciplinary boundaries: the firmware-to-RF interface, the RF-to-FPGA interface, the FPGA-to-data-pipeline interface, and the data-to-model interface, plus a closed-loop intercept-and-act path that any single-domain implementation would require external infrastructure to reproduce. Each requires cross-disciplinary reasoning to characterize and resolve.

The specific problem being solved is not a shortage of technical resources in any individual domain. Comprehensive textbooks, open-source FPGA IP cores, well-documented ML frameworks, and hardware reference designs exist for every component in this system. The gap is the absence of a curriculum that sequences these resources into an integrated learning experience, identifies the interface problems between domains as first-class subjects of instruction, and provides students with a concrete hardware implementation through which to develop and test their understanding.

The problem extends to the educator as well. An instructor preparing a multidisciplinary course of this kind cannot simply draw on deep expertise in a single domain—they must

understand how each discipline's design decisions affect the others well enough to write labs, anticipate failure modes, and evaluate student work at the integration points. In practice, few faculty have equivalent depth across FPGA digital design, RF communications, embedded firmware, and machine learning simultaneously; most come to one or two of these domains as relative outsiders. Without a pre-existing framework that makes the cross-domain interfaces explicit—specifying what the integration problems are, why they arise, and how they can be observed on real hardware—the course preparation burden alone may be enough to prevent such a curriculum from being developed at all. The framework proposed in this paper addresses both sides of this problem: it provides students with the integrated learning experience they currently lack, and it provides educators with the structured reference they would need to deliver it.

III. Proposed Integrated Educational Framework

The proposed solution is a five-module curriculum paired with the benchtop hardware platform described in Section II. The framework assumes foundational domain knowledge and targets the integration layer: the design decisions, interface protocols, and system-level reasoning skills that emerge only when students work across disciplinary boundaries simultaneously.

A. Hardware Platform

The platform was designed around three subsystem layers, with each hardware selection driven by what it asks students to think about rather than by peak performance.

The target layer consists of two STM32 Nucleo-L476RG microcontrollers, each paired with an RFM95W LoRa transceiver operating at 906.5 MHz in the U.S. 915 MHz ISM band. Both targets are TX+RX-capable: Target A initiates DATA, Target B replies with ACK, and both listen continuously for PAUSE commands addressed to their respective Beacon IDs. LoRa was chosen as the RF modality because its chirp spread spectrum waveform exposes a small set of directly tunable parameters—spreading factor, bandwidth, and coding rate—whose effects on link behavior are large enough to observe with benchtop equipment. A student who changes the spreading factor from SF7 to SF12 gains roughly 7.5 dB of link margin per step but cuts the data rate in half [12]; that single decision ripples downstream into FPGA buffer sizing, dataset collection time, and ultimately the variance of the ML training set. This chain of consequences, spanning all four disciplines, is exactly the kind of reasoning the framework is designed to develop. The STM32 platform supports this goal by providing enough peripheral flexibility—multiple independent SPI, I2C, and USART interfaces with DMA support—that students can make and measure the impact of real firmware design decisions rather than working around hardware limitations [10], [13]. Prior work confirms that the RF signatures produced by LoRa hardware are rich enough to support device identification [6] and neural network demodulation [8] at the analysis layer, validating LoRa as a suitable signal source for the full pipeline.

The interceptor (Threat) layer uses a third STM32+RFM95W node, hardware-symmetric with the two targets but operated under a different firmware role. The Threat passively captures every demodulable frame on the shared channel, maintains a per-Beacon-ID RSSI observation table, and—while its onboard user button is held—transmits a PAUSE packet at a 100 ms cadence addressed to whichever non-stale Beacon ID currently has the highest RSSI. Each PAUSE carries a 500 ms hold-off duration in its payload, so a target receiving a single PAUSE remains paused for 500 ms even if subsequent PAUSEs are missed; this gives the protocol roughly five-fold redundancy at the chosen cadence. The PAUSE packet uses the same 32-byte radio frame format as DATA and ACK (Section III-B), so its existence does not require any change to the FPGA ingestion pipeline or the dataset schema. The Threat's hardware symmetry with the targets is itself a teaching point: a single-domain RF receiver design would have no reason to be symmetric with the transmitters, but the framework's intercept-and-act loop requires the third radio to be capable of both reception and well-formed transmission on the same channel.

At the processing layer, the Digilent Nexys A7-100T serves as the FPGA development platform. Its Artix-7 fabric—240 DSP slices, approximately 4,860 Kb of block RAM, and 63,400 LUTs [11]—is large enough to implement a complete signal processing datapath but constrained enough that students must make deliberate resource allocation decisions. This is an important design choice: a platform with unlimited resources would hide the tradeoffs that make FPGA design interesting. Students learn to partition logic between the serial ingestion front-end, the packet parsing and CRC pipeline, and the feature extraction back-end within a fixed budget, which is the same kind of reasoning required in professional FPGA development. The board connects to the STM32 receiver through its PMOD headers, giving students a concrete physical interface to characterize and debug. Prior work has shown that inference pipelines of comparable complexity fit well within the resource envelope of similar devices [1], [3], confirming that the Artix-7 is not a limiting factor for the scope of this curriculum.

The RFM95W was selected after a weighted trade study against three alternative sub-GHz transceiver modules, summarized in Table I. Selection criteria were weighted toward interface compatibility (20%), frequency band (15%), modulation flexibility (15%), documentation quality (15%), cost (10%), power (10%), on-chip packet engine features (10%), and form factor (5%). The RFM95W won on every dimension that matters pedagogically: it exposes a 3.3 V SPI interface the Nucleo can drive directly, operates in the 915 MHz U.S. ISM band, supports both LoRa and FSK/OOK for multi-modulation classification exercises in later modules, and ships as a breadboard-ready breakout with extensive documentation. The EByte E32-915T20D was rejected because its UART-only interface hides the register-level radio programming that

TABLE I
RF TRANSCEIVER TRADE STUDY SUMMARY.

Module	Chipset	Interface	Outcome
Adafruit RFM95W	SX1276	SPI, 3.3 V	**Selected**
AI-Thinker Ra-02	SX1278	SPI, 433 MHz	Rejected: band
EByte E32-915T20D	SX1276	UART only	Rejected: abstraction
HopeRF RFM96W	SX1276	SMD, no brk.	Rejected: form factor

Module 1 is designed to teach; the AI-Thinker Ra-02 was rejected for operating at 433 MHz outside the U.S. ISM band; and the HopeRF RFM96W was rejected for requiring a custom breakout.

At the analysis layer, a host machine accumulates the labeled feature vectors exported by the FPGA and serves as the ML development environment. Students select and train classifiers against this hardware-collected dataset, guided by prior comparative results on RF modulation classification [4] and LoRa-specific deep learning benchmarks [5] as reference points for evaluating their own models.

B. Cross-Domain Frame and Feature Formats

The four cross-domain interfaces identified in Section II are made concrete by three fixed-length data structures that every student implements: a 32-byte radio packet (DATA, ACK, or PAUSE) emitted by the STM32 firmware, a 43-byte forwarding frame wrapped around it by the STM32 interceptor before handoff to the FPGA, and a 32-byte feature vector produced by the FPGA and consumed by the downstream machine learning classifier. These artifacts are the primary instructional objects of the framework—every field in each one corresponds to a design decision made in one disciplinary domain and observed in another.

Table II shows the radio packet. The metadata fields (Beacon ID, TX power, modulation code, spreading factor, bandwidth code) serve a dual role: they give the downstream ML pipeline supervised ground-truth labels, and they let the FPGA verify its own detection logic against values the transmitting station claims it sent. The single `Modulation Code` byte at offset 6 is a packed field: the low two bits encode the packet type (`00` DATA, `01` ACK, `10` PAUSE, `11` reserved), and the upper six bits encode the modulation kind shifted left by two (LoRa = `0x01` $\ll$ 2, FSK = `0x02` $\ll$ 2, OOK = `0x03` $\ll$ 2). For DATA and ACK packets the 17-byte payload region carries either a pseudo-random sequence for bit-error-rate characterization or a structured pattern that exercises specific demodulator behavior; for PAUSE packets, payload byte 0 carries the target Beacon ID being paused and payload byte 1 carries the hold-off duration in 100 ms units. The trailing CRC-16/CCITT-FALSE enables the receiver to distinguish link errors from downstream corruption.

Table III shows the interceptor-to-FPGA forwarding frame. The interceptor prepends a `0x7E` start delimiter, a length byte, an RX timestamp, and three SX1276-derived measurement fields (RSSI, SNR, frequency error) that are not available to the originating station. The full 32-byte radio packet is carried verbatim as the payload, and a frame-level CRC-8 protects the transport layer between the STM32 interceptor and the FPGA independently of the inner LoRa CRC. At 115200 8N1 the 43-byte frame plus framing overhead yields a steady-state capacity of approximately 28 frames per second, which comfortably supports the two-target ping-pong default configuration plus interleaved PAUSE captures and scales by a factor of three before SPI handoff becomes necessary in Module 2.

TABLE II
RADIO PACKET FORMAT (32 BYTES; SAME LAYOUT FOR DATA, ACK, AND PAUSE).

Field	Offset	Size
Preamble sync (`0xAE5A`)	0	2 B
Beacon ID (sender)	2	1 B
Sequence number	3	2 B
TX power (dBm)	5	1 B
Modulation code (kind$\ll$2 | type)	6	1 B
Spreading factor	7	1 B
Bandwidth code	8	1 B
Timestamp (ms)	9	4 B
Payload (DATA/ACK) or PAUSE args	13	17 B
CRC-16/CCITT-FALSE	30	2 B

TABLE III
INTERCEPTOR-TO-FPGA FORWARDING FRAME (43 BYTES).

Field	Offset	Size
Start delimiter (`0x7E`)	0	1 B
Frame length	1	1 B
RX timestamp (ms)	2	4 B
RSSI (dBm, two's comp.)	6	1 B
SNR (0.25 dB, signed)	7	1 B
Frequency error (Hz, int16)	8	2 B
Radio packet (Table II)	10	32 B
Frame CRC-8	42	1 B

The third cross-domain artifact is the feature vector produced by the FPGA ingestion pipeline and consumed by the downstream machine learning classifier. Table IV shows its layout: a 32-byte row aligned to a single block-RAM entry, indexed by Beacon ID in a 256-row dual-port memory. Three of its fields (inter-arrival time, RSSI delta, priority score) are computed on the FPGA from per-station state registers rather than copied from the forwarding frame, which means the feature vector is not a passive repackaging of received data but the first point at which the FPGA begins to abstract away raw packet contents in favor of classifier-ready features. The `pkt_type` field at offset 3 is extracted from the low two bits of the radio packet's `Modulation Code` byte; the `target_id_if_pause` and `pause_duration_units` fields at offsets 28–29 are valid only when `pkt_type` = PAUSE and are otherwise zero. The format is fixed at module boundary specification time precisely so that Module 2 (FPGA feature extraction) and Module 4 (host-side classifier training) can be developed against the same contract without either team blocking on the other—the same style of interface-first design that the

TABLE IV
FEATURE VECTOR FORMAT (32 BYTES, ONE BRAM ROW).

Field	Offset	Size
Beacon ID	0	1 B
Modulation code (kind only)	1	1 B
Spreading factor	2	1 B
`pkt_type` (DATA/ACK/PAUSE)	3	1 B
RSSI (latest)	4	2 B
SNR (latest)	6	2 B
Frame count	8	4 B
Last arrival timestamp	12	4 B
Inter-arrival time	16	4 B
RSSI delta	20	4 B
Priority score	24	4 B
`target_id_if_pause`	28	1 B
`pause_duration_units`	29	1 B
Reserved / padding	30	2 B

framework teaches with the radio and forwarding frames, now applied to the FPGA-to-ML boundary.

A subtle but critical fourth interface is the clock-domain boundary between the STM32 interceptor's asynchronous UART output and the FPGA's 100 MHz system clock. The design crosses this boundary with a dual-flip-flop synchronizer on the `uart_rxd` input—the simplest sufficient CDC structure for a single-bit asynchronous signal—and every other signal in the ingestion pipeline is clocked by the same system clock. This is the one intentional clock-domain crossing in the Module 2 design, and it is deliberately isolated so that students can reason about metastability in exactly one place rather than as an unbounded global concern. Without this discipline, the RF-to-FPGA interface would become a source of intermittent, difficult-to-reproduce failures that would obscure the higher-level integration problems the framework is designed to surface.

The separation of concerns between these frame and feature formats is itself a teaching point: the radio packet is the object the originating firmware owns and the ML pipeline labels against, while the forwarding frame is the object the interceptor firmware and the FPGA ingestion datapath own. A student who changes the radio packet layout without coordinating the interceptor's payload offset produces exactly the kind of silent cross-domain failure the framework is designed to surface.

C. Curriculum Module Sequence

Module 1: RF Hardware and Subsystem Design. Students implement bidirectional cooperative-target firmware on the STM32 platform (Target A initiates DATA, Target B replies with ACK, both listen for PAUSE), implement the interceptor firmware with its per-ID RSSI table and button-gated PAUSE scheduler, design the packet protocol and transmission scheduling algorithm, and validate the physical interceptor-to-FPGA serial interface. The primary learning objective is understanding how firmware design choices—SPI clock rate, DMA configuration, interrupt latency, half-duplex TX/RX role switching—affect the timing and integrity of the data stream delivered to the FPGA. The Nucleo-L476RG's four independent serial interfaces and hardware-accelerated peripherals [10] provide enough configuration space to make these tradeoffs concrete and experimentally observable.

Module 2: FPGA RTL Design and Simulation. Students implement the FPGA-side data ingestion pipeline in VHDL or Verilog: serial receiver, packet parser, CRC checker, and a feature extraction datapath. Simulation testbenches use recorded captures from Module 1 (DATA, ACK, and PAUSE traffic interleaved) as stimulus. Students encounter FPGA-specific design constraints—timing closure, clock domain crossing, memory resource allocation—in the context of a real signal protocol rather than a synthetic exercise. Cloud-based FPGA toolchains have demonstrated that structured, hardware-grounded lab sequences can make this learning accessible without requiring specialized hardware beyond a web browser [9]; this module follows that model while adding the RF protocol context that a standalone digital design course omits.

Module 3: System Integration and Data Collection. Students connect the hardware subsystems built in Modules 1 and 2 and characterize end-to-end data flow: from RF transmission through interception and FPGA processing to feature output. Latency, throughput, and packet error rate are measured. Synchronization mismatches and protocol errors that were invisible in per-subsystem development surface here as integration failures that must be diagnosed and resolved. Students then execute a structured data collection campaign across multiple operating configurations, producing a labeled RF dataset. This module deliberately surfaces the distribution shift problems that arise when training data is drawn from real hardware under one operating regime (single-station clean traffic) but evaluated under another (multi-station contended traffic with interleaved PAUSE), generalizing the well-known sim-to-real distribution-shift result [6] to a fully on-hardware setting.

Module 4: AI-Driven Classification Pipeline. Using the hardware-collected dataset from Module 3, students preprocess FPGA-extracted features, train and evaluate classifiers, and analyze classification accuracy as a function of SNR and number of concurrent stations. Streaming inference architectures [3] and FPGA-accelerated ML pipelines [2] serve as reference designs for students considering on-device deployment. Comparative architecture studies [4] and LoRa-specific classification results [5], [6], [8] provide empirical benchmarks against which students evaluate their own model performance.

Module 5: System Validation and Benchmarking. Students stress-test the complete pipeline along four dimensions (station density, rapid parameter change, low-SNR operation, and extended duration), measure total system response time from RF transmission to classification decision via a six-stage GPIO-toggle decomposition, characterize the closed-loop intercept-and-act behavior of the PAUSE protocol (response latency from Threat-PAUSE TX to Target `"I'm Jammed"` log entry; recovery latency after button release; RSSI-targeting accuracy with two Targets at different distances), and assemble a cross-domain failure-mode catalog whose entries trace observed failures back to specific design decisions in earlier modules.

The benchmarking exercise requires active engagement across all four domains: RF link budget analysis, FPGA timing and resource reports, serial interface throughput, and ML classification accuracy curves. The module's culminating artifacts are an empirically derived system specification table, the closed-loop PAUSE behavioral measurement set, and a structured failure-mode catalog whose chain-depth distribution measures the framework's pedagogical interconnection.

D. Pedagogical Rationale

The framework's central design principle is that the interfaces between disciplinary domains are the primary teaching objects, and this principle gives it concrete advantages over existing single-domain curricula. Prior work has demonstrated that hardware-grounded, progressive lab sequences are effective within a single discipline [9], [10]; this framework extends that validated model to four disciplines simultaneously. The key difference is that integration failures—a firmware DMA timing error that corrupts the FPGA data stream, an RSSI-table staleness threshold that causes the Threat to pick the wrong target, or a station's TX power change that degrades classifier accuracy—are not obstacles to be avoided but structured learning events. Each one forces the student to reason across domain boundaries in a way that no isolated course can replicate.

The framework is also designed to produce a category of evidence that purely accuracy-driven approaches do not. Prior work on LoRa signal classification consistently reports peak accuracy on curated datasets [5], [6], [8] but does not report cross-module regression behavior, integration-failure root-cause depth, or closed-loop intercept-and-act latency—because those metrics presuppose an integrated system that the cited works are not designed to produce. By instrumenting every module with measurement scaffolds whose outputs are consumed by the next module, and by requiring the Module 5 stress and benchmarking deliverables to demonstrate cross-module survival rather than module-local accuracy, the framework makes integration the graded artifact rather than a side effect.

Because the framework is intended to be adopted by instructors who may not share the author's depth in all four domains, the assessment structure is specified as concretely as the hardware. Table V shows the Module 1 assessment breakdown as a representative example: each per-session deliverable maps one-to-one onto a cross-domain interface from Section II, and the weights reflect the difficulty of the integration work rather than the volume of code produced. The trade study and hardware requirements document together account for 40% of the module grade even though they contain no executable code, because the framework treats the *justification* of a cross-domain design decision as a first-class graded artifact. This rubric structure is reproduced with analogous weights in each of the remaining four modules.

Finally, the framework is designed for reproducibility and transferability. All components—the Nexys A7-100T, the STM32 Nucleo-L476RG, and the RFM95W—are commercially available. Prior work has validated that the core technical pipelines are feasible within these hardware resource budgets [1], [3], [7], meaning an instructor at another institution can deploy the same curriculum without custom hardware or proprietary infrastructure. The framework's contribution is not a novel algorithm or hardware design but a structured, teachable model of how these four engineering disciplines connect at the system level—with each connection made concrete through implementation on real hardware.

TABLE V
MODULE 1 ASSESSMENT RUBRIC (REPRESENTATIVE).

Deliverable	Weight	Due
RF module trade study report	20%	Session 1
Hardware requirements document	20%	Session 2
Cooperative target subsystem design	30%	Session 3
Interceptor subsystem and end-to-end	30%	Session 4

IV. NUMERICAL RESULTS AND ANALYSIS

This section presents the *expected* quantitative behavior of the framework across all five instantiated modules, derived from closed-form analysis, vendor datasheet parameters, and framework-defined target counts. Each subsection pairs the expected envelope with explicit scaffolding—placeholder columns, acceptance bounds, and citations to the lab handout that produces the measurement—so that, as student cohorts execute the curriculum, the actual measured values can be filled into the same tables and compared against the expectations stated here. No measured values are reproduced in this section; every numeric entry is either a derived expectation or a target threshold. For Module 1 (RF Hardware and Subsystem Design), a closed-form link-budget analysis establishes the physical-layer operating envelope students encounter in Lab Handouts 1-D through 1-G. For Module 2 (FPGA RTL Design and Simulation), the full ingestion pipeline resource budget and an eight-scenario verification suite are tabulated. For Module 3 (System Integration and Data Collection), the end-to-end latency budget, throughput and FIFO headroom envelope, and data collection campaign volume are derived from the Lab Handout 3-D through 3-H specifications. For Module 4 (AI-Driven Classification Pipeline), the dataset audit and task-specific feature selection, the four-classifier-family architectures with Wilson-interval statistical reporting, the operating-condition shift definition with fine-tuning mitigation, and the multi-task real-time inference latency budget are derived from the Lab Handouts 4-A through 4-H specifications. For Module 5 (System Validation and Benchmarking), a four-dimensional stress-test matrix, a six-stage end-to-end latency decomposition with explicit uncertainty propagation, a six-point benchmarking matrix with baseline-regression rules, the system specification table to be populated from measured data, the closed-loop PAUSE behavioral measurement set, and a chain-depth-tagged failure-mode catalog are derived from the Lab Handouts 5-A through 5-E specifications.

TABLE VI

MODULE 1 EXPECTED LINK BUDGET AT $d = 1$ M, $f_c = 906.5$ MHZ, $B = 125$ KHZ, $P_{\mathrm{tx}} = +14$ DBM. THE "MEASURED $\overline{\mathrm{RSSI}}$" COLUMN IS LEFT BLANK TO BE FILLED FROM THE LH1-D 100-PACKET CAPTURE; A MEASURED VALUE WITHIN ± 6 DB OF -21.6 DBM PASSES THE MODULE 1 ACCEPTANCE CRITERION.

SF	Sens. (dBm)	P_{rx} (dBm)	Margin (dB)	Meas. $\overline{\mathrm{RSSI}}$ (dBm)
SF7	-123	-21.6	101.4	_____
SF8	-126	-21.6	104.4	_____
SF9	-129	-21.6	107.4	_____
SF10	-132	-21.6	110.4	_____
SF11	-133	-21.6	111.4	_____
SF12	-136	-21.6	114.4	_____

A. RF Link Budget and Expected RSSI

The Module 1 baseline configuration (Lab Handout 1-D) operates the SX1276-based RFM95W at $f_c = 906.5$ MHz, spreading factor SF7, bandwidth $B = 125$ kHz, coding rate 4/5, and transmit power $P_{\mathrm{tx}} = +14$ dBm into a 915 MHz helical spring antenna. The free-space path loss for isotropic radiators is

$$L_{\mathrm{FSPL}}(d) = 20 \log_{10}\left(\frac{4\pi d f_c}{c}\right), \tag{1}$$

which at $d = 1$ m evaluates to 31.59 dB at 906.5 MHz. Assuming realistic on-breadboard spring-antenna efficiency losses of $G_{\mathrm{tx}} = G_{\mathrm{rx}} \approx -2$ dB and no additional obstruction, the expected received power is

$$P_{\mathrm{rx}} = P_{\mathrm{tx}} + G_{\mathrm{tx}} + G_{\mathrm{rx}} - L_{\mathrm{FSPL}}(1\,\mathrm{m}) \approx -21.6 \text{ dBm}. \tag{2}$$

Table VI tabulates the expected link budget against the SX1276 LoRa sensitivity floors for representative spreading factors at $B = 125$ kHz [12], with a placeholder column for the mean RSSI a student cohort will measure during LH1-D. The SF7 operating point is expected to leave $\sim$101 dB of link margin at 1 m bench separation, and even the longest configured spreading factor (SF12) is expected to retain the same margin plus an additional 12.5 dB of processing gain [5], [12]. This wide expected margin is what makes SF7 the correct pedagogical default: students should observe clean packet reception and valid CRCs without a range environment, while still being able to reduce power or increase distance in Module 5 to recreate marginal-link behavior without changing the protocol stack.

The Module 1 acceptance criterion ($\geq$95% packet reception, mean SNR $>$+5 dB, and measured mean RSSI within ± 6 dB of (1)) translates this expected envelope into a concrete, testable outcome for every student lab group. A measured deviation outside the ± 6 dB window forces the student to reason across disciplinary boundaries—soldered antenna joint, breadboard parasitics, or firmware PA configuration—which is the cross-domain diagnosis skill the framework exists to develop.

TABLE VII

MODULE 2 PIPELINE EXPECTED RESOURCE BUDGET ON `xc7a100tcsg324-1`. THE BOTTOM ROW HAS A BLANK "MEASURED" FIELD FOR THE COHORT'S VIVADO UTILIZATION SUMMARY.

Block	LUTs	FFs	BRAM
Dual-FF synchronizer	2	2	0
UART receiver	80	50	0
Receive FIFO (64×8)	20	15	0
Packet parser FSM	200	300	0
CRC-8 engine	30	10	0
CRC-16 engine	50	20	0
Feature extraction	150	200	0
Per-station state RAM	0	0	1
Feature BRAM	0	0	4
Host UART export	60	40	0
Expected total	**600**	**640**	**5**
Measured total	_____	_____	_____
Device available	63,400	126,800	270
Expected util.	$<$1%	$<$1%	$\sim$2%

B. Module 2 Pipeline Resource, Timing, and Verification Coverage

The Module 1 FPGA ingestion deliverable (Lab Handout 1-H) is a deliberately minimal four-block bring-up skeleton: a 115200 8N1 UART receiver, a 16×8 byte FIFO, a frame-detector FSM synchronized on the `0x7E` delimiter, and a seven-segment display driver. The full pipeline that the framework targets—including dual-flip-flop synchronizer, UART receiver, byte FIFO, packet parser FSM, CRC-8 transport check, CRC-16 payload check, feature extraction datapath, per-station state RAM, feature BRAM, and host UART export—is specified in Module 2. Quantifying this full pipeline is important because it is the state in which the platform enters Module 4 (machine learning), and any resource or timing headroom claim about the framework must be made at that scale, not at the Module 1 skeleton scale.

Table VIII shows the expected block-level resource budget for the complete Module 2 pipeline targeting `xc7a100tcsg324-1`, with a measured-totals column for the student's post-implementation report. Expected totals are approximately 600 LUTs, 640 flip-flops, and five 18 Kb BRAM tiles. The two largest BRAM consumers are the 256-row per-station state memory and the 256-row $\times$ 256-bit feature storage, both indexed directly by Beacon ID so that no hashing or associative lookup is required. Against device totals of 63,400 LUTs, 126,800 flip-flops, and 270 BRAM tiles, aggregate expected utilization remains under 1% of logic and approximately 2% of block RAM—leaving essentially the entire device available for the Module 4 inference engine.

Timing closure at the target 100 MHz system clock is expected with worst negative slack greater than 3 ns in the reference implementation, and dynamic power dissipation is expected below 200 mW; both quantities are to be recorded from the student's Vivado timing and power reports against the expected envelope above. The critical path is expected to fall on the CRC-16 XOR tree; if it does not close on a given student's implementation, the standard remediation is

TABLE VIII

MODULE 2 VERIFICATION CHAIN. "REFERENCE RUN" IS THE FRAMEWORK'S OWN IMPLEMENTATION; THE DIFF COLUMN IS THE BYTE-DIFFERENCE COUNT BETWEEN HARDWARE DUMP AND PYTHON REFERENCE FOR LH2-G OR PASS/FAIL FOR UNIT TESTBENCHES.

#	Testbench	Coverage target	Reference run	DIFF
1	`tb_uart_rx`	Single-byte 0xA5 recovery, generics	PASS	—
2	`tb_uart_rx_regression`	62,952-byte replay vs. `.bin`	PASS, 0 mismatches	0-byte diff
3	`tb_byte_fifo`	16-write/16-read, registered output	PASS	—
4	`tb_byte_fifo_overflow`	65 writes, full-flag latch, reset clear	PASS	—
5	`tb_pkt_parser`	3 frames: VALID, REJECT-len, VALID	PASS	—
6	`tb_crc`	CRC-8/CRC-16 "123456789" golden vector	PASS	—
7	`tb_feature_extract`	6 frames, 3-row read-modify-write	PASS	—
8	`tb_ingest_top` (clean)	Full pipeline, single-station capture	PASS	0-byte diff
9	`tb_ingest_top` (mixed)	Full pipeline, ping-pong + button-hold	PASS	0-byte diff
10	`tb_ingest_top` (corrupt)	Full pipeline, ∼5% bit-flipped frames	PASS	0-byte diff

TABLE IX

MODULE 3 EXPECTED END-TO-END LATENCY FOR A 32-BYTE RADIO PAYLOAD AT $B = 125$ KHZ, CR=4/5. THE "MEASURED" COLUMN IS BLANK FOR THE LH3-D GPIO-TOGGLE CAPTURE.

SF	L_{RF} (ms)	L_{e2e} exp. (ms)	Meas. (ms)	RF share
SF7	∼50	∼55	_____	∼91%
SF10	∼370	∼375	_____	∼99%
SF12	∼1503	∼1505	_____	∼99.7%

a single pipeline register splitting the XOR network across two cycles, which reclaims several nanoseconds of slack at no resource cost. The only intentional clock-domain crossing in the design is the dual-flip-flop synchronizer on `uart_rxd`, isolating metastability to a single well-defined structure.

Verification coverage is specified as a tiered eight-testbench regression chain that every Module 2 submission must pass. Table IX summarizes the testbenches and their results from the framework's reference implementation. The chain deliberately pairs unit-level coverage (one testbench per replaced or newly-added block from LH2-B through LH2-F) with full-pipeline coverage (LH2-G's `tb_ingest_top` replays a recorded interceptor capture through every stage, byte-for-byte). The corrupted-stream variant of LH2-G is the strongest single check: an external Python reference and the VHDL hardware independently parse, CRC-validate, and accumulate the same bit-flipped capture, and their per-Beacon-ID feature-vector dumps are required to agree byte-for-byte across all 256 BRAM rows. The existence of this fixed regression chain also gives instructors at other institutions an objective basis for evaluating reproducibility: two independent implementations of the framework can be compared by running both against the same stimulus files and `diff`-ing the resulting feature-vector dumps. In the reference implementation, all eight testbenches passed in the order shown; the LH2-G reference dumps matched the Python reference with zero byte differences across all three captures.

Three notable curriculum-tightening contributions emerged from running this verification chain on the framework's reference hardware. (i) Module 1's USB-to-UART adapter requirement for capturing the Threat's forwarding-frame stream was eliminated by adding a one-line passthrough that routes the FPGA's incoming PMOD UART pin directly to the Nexys A7's onboard FT2232H USB-UART output (`UART_RXD_OUT`, pin `D4`); the same USB cable that programs the FPGA then exposes the capture stream to the host as a virtual COM port, removing a piece of hardware students would otherwise need to source. (ii) The interceptor (Threat) firmware exhibited a dropped-frame rate of effectively 100% (one frame forwarded per board reset, then `tx_busy` stuck high) until `USART3_IRQn` was enabled at the NVIC; without that interrupt, `HAL_UART_Transmit_DMA`'s Transmission-Complete callback never fired and the busy flag was never cleared. The corresponding CubeMX configuration step was missing from the original LH1-C handout and has been added. (iii) `feature_bram` required an explicit synchronous reset clearing the storage array and the registered read outputs; the VHDL signal-declaration default `(others => (others => '0'))` alone produced stale read values in Vivado xsim during reset until an actual write to the read address. These three artifacts are documented in the framework's TestLog and ExplanationsLog and are propagated into the published handouts so that subsequent cohorts encounter the corrected procedure rather than rediscovering the same three pitfalls.

C. Module 3 End-to-End Latency Budget

The end-to-end latency L_{e2e} measured in Lab Handout 3-D is the elapsed time from radio TX start to feature-vector availability in the FPGA feature BRAM, decomposed as

$$L_{\mathrm{e2e}} = L_{\mathrm{RF}} + L_{\mathrm{RX_FW}} + L_{\mathrm{UART}} + L_{\mathrm{FPGA}}. \tag{3}$$

The UART contribution at 115,200 8N1 over the 43-byte forwarding frame of Section III-B is fixed:

$$L_{\mathrm{UART}} = \frac{43 \times 10}{115{,}200} \approx 3.73 \text{ ms}. \tag{4}$$

The FPGA pipeline contribution is bounded above by six clock cycles of the 100 MHz ingestion datapath after the last byte of the frame, well under 0.1 ms. The interceptor firmware ISR plus frame construction is expected to add under 1 ms. The RF on-air time dominates and depends strongly on the LoRa spreading factor; for a 32-byte payload at $B = 125$ kHz, CR=4/5, the SX1276 airtime formula [12] yields the expected values in Table X, alongside placeholders for the cohort's GPIO-toggle measurements at LH3-D.

The Lab Handout 3-D acceptance criterion requires the GPIO-toggle measurement at SF7 and SF10 to fall within $\pm 15\%$ of the predicted L_{e2e}, and requires the SF7→SF10 difference (predicted ∼320 ms) to be reproduced within the same tolerance. A deviation outside that window forces the student to localize whether the unexpected latency originates in firmware ISR jitter, in UART baud drift, or in an FPGA timing-closure regression—a diagnosis that necessarily spans three of the four cross-domain interfaces of Section II.

The dominance shift visible in Table X carries an architectural consequence that reappears in the Module 5 benchmarking deliverable: any optimization targeting the FPGA or

TABLE X
MODULE 3 EXPECTED PER-CONFIGURATION STEADY-STATE FRAME RATE AND 64-ENTRY FIFO HEADROOM FOR THE LH3-E STRESS MATRIX. THE "MEAS. FIFO PEAK" COLUMN IS BLANK FOR THE COHORT'S LED-MAPPED PEAK OBSERVATION.

Stations	Mode	Interval	fps	Exp. peak	Meas.
1	Fixed	2 s	0.50	<10%	_____
2	Fixed	2 s / 3 s	0.83	<15%	_____
2	ALOHA	[1,3] s	∼1.0	15–30%	_____
3	Fixed	1 s each	3.0	25–40%	_____
3	ALOHA	[0.5,2] s	3.0–4.0	40–70%	_____

TABLE XI
MODULE 3 CAMPAIGN VOLUME ACROSS LH3-F AND 3-G, WITH EXPECTED EFFECTIVE ROW COUNT UNDER THE ≤5% PER BOUND AND A BLANK MEASURED COLUMN.

Campaign	Sweep	Steps	Pkts	Exp. rows	Meas.
C1	TX power	4	800	≥760	_____
C2	Spreading factor	6	1,200	≥1,140	_____
C3	Modulation type	2	600	≥570	_____
C4	Distance	4	800	≥760	_____
C5	Station count / mode	5	1,500	≥1,200	_____
Total	—	21	**4,900**	**≥2,500**	_____

UART stages produces sub-1% improvement at SF12 and at most a single-digit-percentage improvement at SF7. Latency-critical deployments of this pipeline must reduce the RF on-air time, not the digital processing time. This is a non-obvious system-design conclusion that students reach only after instrumenting and measuring across all four disciplinary domains simultaneously.

D. Throughput, Station Density Scaling, and FIFO Headroom

The UART-imposed throughput ceiling of the Module 3 pipeline is

$$\text{Throughput}_{\text{max}} = \frac{115{,}200}{43 \times 10} \approx 268 \text{ fps}. \tag{5}$$

The achievable rate is far below this ceiling because the LoRa on-air time of Section IV-D caps each station's per-emitter frame rate well below 1 fps at SF7. The Lab Handout 3-E stress test therefore characterizes not the UART ceiling but the FIFO headroom under burst arrivals, which is the only scenario in which the 64-entry receive FIFO can approach overflow.

Table XI summarizes the expected steady-state operating point for each configuration. For the framework's three-radio default, two stations correspond to Target A and Target B in normal ping-pong; the three-station rows are reached by enabling Target B as an independent initiator in addition to the ping-pong responder role, or by including PAUSE traffic from the Threat. The fixed-interval rows are expected to be bounded by deterministic transmission scheduling and stay well clear of FIFO pressure; the ALOHA rows admit collisions and burst arrivals where two or three frames land within a single UART transfer window of approximately 3.7 ms, producing the elevated FIFO peaks shown. The Lab Handout 3-E pass criterion (peak FIFO occupancy <75%, no overflow alarm) is set so that the framework scales to three concurrent stations without redesign. If the criterion fails, the documented mitigation is to increase the FIFO depth from 64 to 256 entries at a cost of one additional 18 Kb BRAM tile, which is negligible against the ∼2% block-RAM utilization of the complete Module 2 pipeline reported in Table VIII.

The framework also separates a debug bitstream (Lab Handouts 3-B through 3-D) from the production bitstream used for data collection (Lab Handouts 3-E through 3-H). The debug build adds an 11-probe ILA with a 1024-sample capture buffer, consuming approximately four additional BRAM tiles; the production build sets the top-level `g_DEBUG` generic to false, prunes the ILA at elaboration, and returns block-RAM usage to the Module 2 baseline of Table VIII. Lab Handout 3-E requires that the production bitstream's measured PER fall within one percentage point of the debug bitstream's PER for the same single-station configuration; any larger discrepancy is treated as evidence that ILA insertion altered timing in the debug build, which is a pedagogically important regression to surface rather than to mask.

E. Data Collection Campaign Volume and Split Integrity

The five single-variable sweep campaigns of Module 3 (Lab Handouts 3-F and 3-G) produce a labeled feature dataset at known target volume. Table XII tabulates the target packet count per campaign, the resulting raw row count, and the expected effective row count after the Module 1 acceptance bound of ≤5% PER (Section IV-H) is applied at each level. A blank "Measured" column records the actual count produced by the cohort's collection runs.

The aggregate target of 4,900 transmitted packets is expected to yield at least 2,500 effective feature vectors after PER attrition, which is the minimum threshold below which the stratified 70/15/15 split of Lab Handout 3-H cannot guarantee at least 30 test-set samples per minority class. This in turn drives the per-level packet-count choices in Lab Handouts 3-F and 3-G: 200 packets per level at C1, C2, and C4, and 300 per level at C3 and C5, are not arbitrary collection durations but the smallest values that satisfy the downstream split constraint at the worst-case PER allowance.

The Lab Handout 3-H merge script keys the stratified train/val/test split on `run_id`—one value per invocation of `collect.py`—rather than on `campaign_id` or on individual rows. Samples within a single run share channel conditions and hardware state and must therefore travel into the same partition; splitting at the row level would leak channel context into the test set, while splitting at the campaign-family level would collapse legitimate distinct runs into a single partition bucket. With 21 collection runs total across C1–C5e, a 70/15/15 partition assigns roughly 15/3/3 runs to train/val/test per stratum (`label_sf`), which is the granularity at which the Lab Handout 3-H verification check (≥30 test samples per class) is evaluated.

The framework also defines a tolerance budget on label integrity. The merge script flags every row where the originating station's self-reported parameters disagree with the campaign-configured ground truth, and Lab Handout 3-H requires the aggregated mismatch rate to fall below 2%. A higher rate triggers a firmware audit rather than silent record discard, because a systematic self-report disagreement is itself a pedagogically valuable cross-domain failure: a station's modulation-code register that did not get reflashed, or an SF override applied to the wrong Beacon ID, are integration faults that manifest only at the data-to-model interface and are exactly the failure mode the framework is designed to surface.

F. Module 1 Acceptance Criteria as Quantitative Outcomes

The Module 1 analyses in Sections IV.A and IV.B are tied together by a set of pass/fail criteria that constitute the Module 1 quantitative deliverable. A student cohort completes Module 1 successfully when, across all lab groups: (i) every assembled RFM95W breakout draws between 0.5 and 5 mA of idle current at 3.3 V and reads `RegVersion` = `0x12`; (ii) the LH1-D bidirectional ping-pong test achieves $\geq$95% reception with zero CRC errors across 100 consecutive DATA/ACK pairs and mean RSSI within $\pm$6 dB of the Table VI prediction; (iii) the LH1-G interceptor demonstrates the per-Beacon-ID RSSI table populated with both Targets and the button-gated PAUSE protocol selecting the highest-RSSI target as the PAUSE recipient, with the targeted station printing `"I'm Jammed"` on its USART2 debug log within one PAUSE cadence (100 ms) of the button being pressed; and (iv) the Nexys A7 end-to-end display updates within 2 seconds of a station being added or removed, with zero Vivado implementation timing violations. Each of these criteria corresponds to one of the four cross-domain interfaces defined in Section II, so a failure at any criterion isolates the affected interface for remediation without ambiguity.

G. Module 3 Acceptance Criteria as Quantitative Outcomes

The Module 3 analyses in Sections IV-D–IV-F are tied together by a parallel set of pass/fail criteria that constitute the Module 3 quantitative deliverable, with each criterion mapped one-to-one onto a cross-domain interface from Section II. A student cohort completes Module 3 successfully when, across all lab groups: (i) the eight-item pre-integration physical-interface checklist of Lab Handout 3-A passes, including a verified common-ground jumper between the STM32 interceptor and the Nexys A7 PMOD JA, and the first live frame of Lab Handout 3-C is captured with parsed Beacon ID, RSSI, and sequence number that match the STM32 interceptor debug log within $\pm$2 dB on RSSI; (ii) the Lab Handout 3-D measurement at SF7 yields mean L_{e2e} within $\pm$15% of the $\sim$55 ms prediction of Table X and the SF7$\rightarrow$SF10 difference reproduces the predicted $\sim$320 ms shift, while the four-stage PER decomposition (RF, UART framing, CRC, FIFO overflow) sums to an aggregate PER $\leq$5%; (iii) all five stress configurations of Lab Handout 3-E run for the full five-minute window without latching the FIFO overflow alarm at peak occupancy below 75%, and the production bitstream's measured PER lies within one percentage point of the debug bitstream's PER for the same single-station configuration; and (iv) the merged dataset produced by Lab Handout 3-H contains $\geq$2,500 feature vectors against the 4,900-packet target of Table XII, with label-mismatch rate below 2%, a stratified 70/15/15 split keyed on `run_id`, and at least 30 test-set samples per minority class on the spreading-factor label. As with the Module 1 criteria, a failure at any one of the four isolates the affected interface—physical, link, throughput, or data-to-model—for remediation without ambiguity.

H. Module 4 Dataset Audit and Task-Specific Feature Selection

The Module 4 deliverable opens with a ten-check data integrity audit (Lab Handout 4-A) that runs before any preprocessing or model training. Each check guards against a specific failure mode that would otherwise propagate into trained classifiers without raising a runtime error: schema drift between the LH3-H contract and the loaded CSV, row-count truncation below 2,500, null-value exceedance over 1% on any feature column, duplicate (`run_id`, `beacon_id`, `obs_count`) tuples, out-of-range values on RSSI/SNR/SF/`mod_code`, undecoded two's-complement signed fields, label-mismatch above the LH3-H 2% threshold, class imbalance below 150 samples per class, `run_id` leakage across train/val/test partitions, and sentinel rows (`obs_count` = 1) where inter-arrival time and RSSI delta are undefined. Any failed check halts Module 4 until the data contract is restored at the source rather than patched in the loader; this discipline is what makes the data contract auditable across module boundaries.

Four classification tasks then proceed in parallel with disjoint feature sets, derived mechanically from the label-leakage table of Table XIII. Each task retains the seven base features [`rssi`, `snr`, `inter_arrival_ms`, `rssi_delta`, `obs_count`, `priority_score`, `distance_m`] minus its task-specific forbidden columns. The new packet-type task (DATA / ACK / PAUSE) is the framework's most aggressive label-leakage trap: `pkt_type` is itself a feature column extracted from the radio packet's `Modulation Code` byte, and any model trained with `pkt_type` retained will trivially achieve perfect accuracy on it. The forbidden list correctly excludes `pkt_type` from the packet-type task's feature set, forcing the model to discriminate DATA, ACK, and PAUSE from RSSI dynamics, inter-arrival patterns, and per-station observation history alone. A leakage sanity check trains a logistic regression using *only* the forbidden columns and verifies test accuracy $\geq$0.95—a positive result confirms the columns are correctly identified as labels-in-disguise, and any task scoring below 0.95 indicates the forbidden list is incomplete and must be revisited before continuing to LH4-C.

The fitted `StandardScaler` from LH4-B is the single shared preprocessing artifact reused by all four classifier families across LH4-C through LH4-F. The scaler is fit on training data only and applied via plain `transform` on validation and test data; this discipline prevents test-set statistics

TABLE XII
MODULE 4 TASK-SPECIFIC FEATURE SELECTION AND LABEL-LEAKAGE FORBIDDEN COLUMNS (LAB HANDOUT 4-B).

Task	Label	Forbidden columns	Retained
SF detection	`label_sf`	`spread_factor`	6 of 7
Modulation	`label_mod`	`mod_code`, `spread_factor`	5 of 7
Emitter ID	`label_beacon`	`beacon_id`	6 of 7
Packet type	`label_pkt`	`pkt_type`	6 of 7

TABLE XIII
MODULE 4 CLASSIFIER FAMILY ARCHITECTURES AND HYPERPARAMETER SPACES.

Family	Architecture / Grid	Configs
DummyClassifier	majority class	1
Decision tree	`max_depth`=5	1
Random forest	$n_{\text{est}} \in \{100, 300, 600\}$, $d_{\max} \in \{\infty, 10, 20\}$, leaf $\in \{1, 3\}$, feat $\in$ {sqrt,log2}	36
MLP	$F \to 64 \to 32 \to C$, drop=0.3	seed-only
CNN-1D	$8 \times F$ window, Conv16(k3)+ Conv32(k3)+GAP+FC32+drop+FCC	seed-only

from leaking into the model's input distribution. Tree-based classifiers ignore monotonic feature transformations and gain no accuracy from scaling, but the framework still loads scaled matrices everywhere so that all four model families consume identical inputs and can be compared without preprocessing-induced confounds.

I. Module 4 Classifier Architectures, Statistical Reporting, and Operating-Condition Shift

Module 4 trains four classifier families per task, summarized in Table XIV. The grid for the random forest spans $3 \times 3 \times 2 \times 2 = 36$ hyperparameter configurations evaluated with 5-fold stratified cross-validation over the concatenated train+validation split; this concatenation is necessary because at the framework's expected ∼2,500-row dataset size, the bare 15% validation split holds only ∼375 rows, which is too small a sample for reliable hyperparameter selection. The MLP and CNN-1D are trained with Adam at learning rate 10^{-3} and batch size 32, with early stopping on the validation cross-entropy loss at a patience of 15 epochs.

Test accuracy at the framework's expected test-set size requires interval estimation, not point estimates. Module 4 reports a 95% Wilson score confidence interval for every accuracy $\hat{p} = N_{\text{correct}}/n$:

$$\text{CI}_{95}(\hat{p}, n) = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \tag{6}$$

with $z = 1.96$. The Wilson interval is preferred over the standard-normal approximation because it remains in $[0, 1]$ for small n or extreme $\hat{p}$. At $n = 375$ a 2-percentage-point absolute accuracy gap between two models is approximately the threshold below which the Wilson intervals overlap and the difference cannot be claimed as significant. The MLP additionally reports test-accuracy mean ± standard deviation over four random-seed runs (seeds 0–3); a standard deviation exceeding 0.02 on any task flags the model as unstable for deployment regardless of its mean accuracy.

The capstone Module 4 result is the operating-condition shift experiment of Lab Handout 4-F. Every row in the merged dataset is captured under one of two regimes: a *clean* regime drawn from the single-station campaigns (C1 TX-power sweep, C2 spreading-factor sweep, C3 modulation-type sweep, C4 distance sweep), and a *contended* regime drawn from C5 (multi-station ALOHA with collisions and interleaved PAUSE traffic). With the CNN-1D as the chosen architecture (because it consumes 8-row sequences and therefore amortizes single-packet noise over a window of consecutive observations from the same station), the experiment measures three quantities per task:

$$\Delta = \text{Acc}(\text{clean} \to \text{clean}) - \text{Acc}(\text{clean} \to \text{contended}), \tag{7}$$

where the first term trains and evaluates on held-out rows from the clean regime and the second trains on the same clean rows while evaluating on a held-out subset of the contended regime. The fair-comparison rule fits the `StandardScaler` on the training source and applies that same scaler to the evaluation set, treating preprocessing as part of the trained model rather than as a separable post-hoc transform. A non-zero Δ is not framework failure—it is the framework's central pedagogical observation about distribution shift, performed entirely on hardware data without any simulator dependency, providing the empirical foundation for the LH3-H distribution shift analysis that motivated the multi-regime data collection investment in Module 3 in the first place. Mitigation via fine-tuning freezes the CNN backbone trained on clean data and re-trains only the final linear layer on 10–20% of the contended training subset; expected outcome is recovery of at least half of Δ on the SF detection task, demonstrating that the convolutional features generalize across the clean-to-contended boundary while the classifier head was miscalibrated to the clean-only feature distribution.

The robustness of every model is then characterized in Lab Handout 4-G by stratifying the test set along three operating-condition axes: signal-to-noise ratio (5 quantile bins covering the empirical SNR range), station-to-interceptor distance (4 discrete levels at 1, 3, 5, and 8 m drawn from Campaign C4), and concurrent station count (5 sub-configurations C5a–C5e at 1/2/2/3/3 active stations under fixed and ALOHA scheduling). Quintile binning produces approximately 75–80 samples per SNR bucket at the expected test-set size, giving each per-bucket accuracy a 95% Wilson half-width on the order of 0.07 at $\hat{p} = 0.85$—loose enough that small between-model differences at individual buckets are not interpretable, but tight enough that consistent patterns across all five buckets separate genuinely robust models from those whose headline accuracy is driven by easy operating points. The framework defines the *worst-bucket accuracy* on the SF task as the deployable robustness metric: a model with high mean accuracy but degraded worst-case bucket accuracy is a worse field-deployment

TABLE XIV
MODULE 4 EXPECTED END-TO-END RESPONSE TIME AT THE SF7 BASELINE, EXTENDING TABLE X THROUGH HOST INFERENCE. THE "MEASURED" COLUMN IS BLANK FOR THE LH4-H 1,000-CALL MICROBENCHMARK AND THE LH3-D GPIO-TOGGLE CAPTURE.

Stage	Exp. (ms)	Meas. (ms)	Source
RF on-air (SF7, 32 B)	∼50	_____	§IV-D
Interceptor firmware ISR	<0.5	_____	§IV-D
UART transport (43 B, 115,200)	3.73	_____	§IV-D
FPGA pipeline	<0.1	_____	§IV.B
Host inference (4 tasks)	<8	_____	LH4-H target
Total end-to-end	<62	_____	—

TABLE XV
MODULE 5 STRESS-TEST MATRIX, TARGET SUBSYSTEM, PRE-DECLARED PASS/FAIL THRESHOLDS, AND A BLANK "OUTCOME" COLUMN FOR THE COHORT'S PASS/FAIL ENTRY.

ID	Dimension	Threshold	Duration
S1	Station density (1–3, fixed/ALOHA)	PER<5%, FIFO<75%	5 min/cfg
S2	Rapid SF change (SF7→10→12)	≤2 transient miscl.	10 min
S3	Low-SNR sweep (+20→+2 dBm)	Acc≥80% above 0 dB SNR	3 min/lvl
S4	Extended duration (2 stations, SF7)	No FIFO ovf, no acc. drift	30 min

candidate than a model with slightly lower mean and higher worst-case, even though a single number cannot capture both.

J. Module 4 Real-Time Multi-Task Inference Latency Budget

The deployed Module 4 inference engine of Lab Handout 4-H runs all four classifiers concurrently on each incoming feature vector from the FPGA's UART export, producing simultaneous SF, modulation, emitter-ID, and packet-type predictions per packet. Table XV extends the Section IV-D Module 3 latency budget through the host-side multi-task inference stage to give the complete end-to-end response time from radio TX start to displayed classification.

The 1-second beacon interval at the SF7 baseline configuration leaves

$$T_{\text{available}} = T_{\text{beacon}} - L_{\text{UART}} \approx 1000 - 3.73 \approx 996 \text{ ms} \quad (8)$$

of host-side budget between consecutive packets, against which the 8 ms multi-task inference target represents under 1% utilization. The framework still imposes the 8 ms target rather than reporting against the full 996 ms because the headroom must accommodate (i) all four task models running serially, (ii) Python interpreter overhead and occasional garbage-collection pauses, and (iii) the latency tail from scikit-learn's internal predict path. The target is verified by a 1,000-call warm microbenchmark that reports the 99th-percentile latency rather than the mean: long-tailed distributions from GC, interpreter warm-up, and OS scheduling jitter can hide budget violations behind a comfortable mean while producing periodic late predictions in the live pipeline.

A schema manifest written by LH4-A and read by LH4-H provides the deployment safety check. The manifest captures the column order and dtype of the validated dataset, and LH4-H asserts that the live FPGA feature-vector schema matches this manifest before starting inference. Any drift—a column added in a later FPGA bitstream revision, a renamed field in the LH3-H merge script—is a deployment blocker rather than a runtime warning, because feature-order drift produces silently wrong predictions rather than a parsing error. This is the same data-contract discipline that LH4-A established between Modules 3 and 4, now enforced at deployment time as well as at training time.

K. Module 4 Acceptance Criteria as Quantitative Outcomes

The Module 4 analyses in Sections IV-I–IV-K are tied together by a parallel set of pass/fail criteria, each mapped one-to-one onto a cross-domain interface from Section II. A student cohort completes Module 4 successfully when, across all lab groups: (i) the Lab Handout 4-A ten-check data audit passes with no waivers, the merged dataset retains ≥2,500 feature vectors with label-mismatch rate below 2%, and the LH4-B leakage sanity check yields ≥0.95 forbidden-column-only test accuracy on all four tasks; (ii) the random-forest grid search beats both the DummyClassifier majority baseline and the depth-5 decision tree on every task with non-overlapping 95% Wilson intervals computed via (7), and the MLP seed-robustness standard deviation stays below 0.02 across four independent seed runs per task; (iii) the Lab Handout 4-F operating-condition shift Δ defined in (8) is computed and tabulated for all four tasks with the fine-tuning mitigation reducing Δ by at least half on the SF detection task, and the LH4-G stratified-accuracy curves are produced for SNR (5 quintile bins), distance (4 levels from C4), and station count (5 sub-configurations from C5); and (iv) the Lab Handout 4-H inference engine loads all four chosen models, the 1,000-call microbenchmark reports a 99th-percentile latency below 8 ms against the budget of Table XV, the schema manifest matches the live FPGA feature-vector layout with no drift, and the five-minute live demonstration produces predictions whose post-hoc accuracy on known station configurations matches the held-out test accuracy within the model's Wilson interval. As with the Module 1 and Module 3 criteria, a failure at any one of the four isolates the affected interface—data contract, statistical reporting, distribution-shift generalization, or real-time deployment—for remediation without ambiguity.

L. Module 5 Stress-Test Matrix and Pre-Declared Thresholds

Module 5 closes the curriculum by characterizing the integrated system rather than its constituent subsystems. The pedagogical principle—explicit in Lab Handout 5-A—is that the stress test *derives* the system specification from empirical evidence rather than testing against an a priori budget. Table XVI summarizes the four-dimensional stress matrix executed across LH5-A and LH5-B, with each dimension targeting a distinct system limitation and a pre-declared pass/fail threshold so that thresholds cannot be retroactively adjusted to make a failing run pass.

The S1 station-density sweep targets the FIFO depth chosen in Module 2 and the UART baud rate chosen in Module 1; the S2 rapid-change test targets the FPGA feature-extraction logic of Module 2 (specifically, the inter-arrival-time computation that crosses an SF transition with a stale prior-packet timestamp); the S3 low-SNR sweep targets the training-data distribution of Module 3 by extending operation into the <0 dB SNR regime that the Module 3 campaigns deliberately undersampled; and the S4 endurance run targets the counter widths and host-side resource hygiene that no shorter test can exercise. Each dimension is therefore not a generic "does it still work" check but a directed probe at a specific upstream design decision, which is the property that makes failure at any Sx row diagnostically informative.

The S1e configuration (3 stations, ALOHA $[0.5, 1.5]$ s) is also the framework's primary use case for the LH3-B/LH3-E debug-vs-production bitstream split. The production bitstream collects the headline PER and FIFO numbers, after which the operator explicitly swaps in the debug bitstream to capture an ILA waveform of the burst, then swaps back before any subsequent throughput or endurance measurement. The bitstream swap is explicit rather than automatic precisely because conflating the two builds was identified as a measurement-validity hazard during Module 3 development.

M. Module 5 End-to-End Latency Decomposition with Uncertainty Propagation

Lab Handout 5-C extends the Module 3 four-stage latency model of Section IV-D into a six-stage decomposition that adds the host-side UART export and Python inference stages, giving a full end-to-end response time from radio TX start to classification print. The decomposition is

$$T_{\text{total}} = T_{\text{RF}} + T_{\text{RX_FW}} + T_{\text{UART_in}} + T_{\text{FPGA}} + T_{\text{UART_out}} + T_{\text{infer}}. \tag{9}$$

Three stages are directly measured (oscilloscope cursor on T_{total}, ILA timestamp on T_{FPGA}, `perf_counter` on T_{infer}); three are calculated from datasheet or framework parameters (T_{RF} from the SX1276 airtime formula [12], $T_{\text{UART_in}} = 3.73$ ms for the 43-byte forwarding frame, $T_{\text{UART_out}} = 3.04$ ms for the 35-byte host export). The interceptor firmware stage $T_{\text{RX_FW}}$ is computed by subtraction and inherits the propagated uncertainty

$$\sigma_{\text{RX_FW}} = \sqrt{\sigma_{\text{total}}^2 + \sum_{i \neq \text{RX_FW}} \sigma_i^2}. \tag{10}$$

A reportable point estimate for $T_{\text{RX_FW}}$ requires that this propagated uncertainty be smaller than the residual itself; otherwise the framework requires the residual to be reported as "not resolvable by subtraction at this instrumentation budget" rather than as a number with a misleadingly tight bound. This rule is what converts a casual stage-by-stage spreadsheet into a defensible specification claim.

Table XVII summarizes the six-stage decomposition at the SF7 baseline configuration with the per-stage uncertainty source. The dominance of T_{RF} at every operating point is the framework's central architectural conclusion: at SF7 the RF stage accounts for roughly 75% of T_{total} and at SF12 for over 99%, so any latency reduction targeting the digital pipeline produces single-digit-percent improvement at SF7 and is essentially invisible at SF12. The actionable consequence—articulated as a writing exercise in Lab Handout 5-D Step 6—is that on-FPGA inference does not reduce T_{total} unless the host PC is eliminated entirely.

TABLE XVI
MODULE 5 SIX-STAGE LATENCY DECOMPOSITION AT THE SF7 BASELINE WITH PER-STAGE UNCERTAINTY SOURCE. THE "MEASURED" COLUMN IS BLANK FOR THE LH5-C SCOPE AND ILA CAPTURES.

Stage	Exp. (ms)	Meas. (ms)	Uncertainty source
T_{RF} (SF7, 32 B)	∼50	____	SX1276 sym. tol., ∼1%
$T_{\text{RX_FW}}$	<0.5	____	Propagated (11)
$T_{\text{UART_in}}$ (43 B)	3.73	____	Baud quant., ∼0.5%
T_{FPGA}	<0.1	____	1 sysclk cycle, 10 ns
$T_{\text{UART_out}}$ (35 B)	3.04	____	Baud quant., ∼0.5%
T_{infer} (4 tasks)	<8	____	LH4-H p99 microbench
Total	<62	____	**Quadrature sum**

TABLE XVII
MODULE 5 BENCHMARKING MATRIX (LH5-D).

ID	SF	Stations	Distance	Power
B1	SF7	1	3 m	+14 dBm
B2	SF10	1	3 m	+14 dBm
B3	SF12	1	3 m	+14 dBm
B4	SF7	2	3 m	+14 dBm
B5	SF7	3	3 m	+14 dBm
B6	SF7	1	8 m	+2 dBm

N. Module 5 Six-Point Benchmarking Matrix and Baseline Regression Rules

Lab Handout 5-D operationalizes the latency model through a six-point benchmarking matrix that varies one axis at a time from the B1 baseline (SF7, 1 station, 3 m, +14 dBm). Table XVIII lists the operating points and the dominant variable each isolates. At each point, 50 T_{total} measurements are captured with the GPIO-toggle setup, and the deliverable is mean, standard deviation, minimum, maximum, and 99th-percentile of T_{total} together with the dominant stage by proportional contribution. Histograms are required at the two extremes, B1 and B3, and the framework predicts that the B1 and B3 distributions have the same shape translated along the time axis by the ∼1450 ms RF airtime difference between SF7 and SF12 for a 32-byte payload.

A regression check against the Module 3 and Module 4 baselines is required at the B4 operating point (2 stations, SF7, 3 m, +14 dBm) for PER, end-to-end latency, classification accuracy on all four tasks, and FIFO peak utilization. The framework imposes seven comparability axes that must be pinned before a regression claim is admissible: spreading factor, station count and scheduling mode, distance, transmit power, target plus interceptor plus FPGA firmware commit hash, bitstream hash, and model artifact hash. A row miss-

TABLE XVIII
MODULE 5 SYSTEM SPECIFICATION SCAFFOLD TO BE POPULATED FROM LH5-A THROUGH LH5-D MEASUREMENTS. EVERY VALUE IS LEFT BLANK IN THIS PAPER; THE COHORT FILLS EACH ROW FROM ITS CITED SOURCE EXPERIMENT.

Parameter	Value	Source
Max. concurrent stations (PER<5%)	_____	LH5-A (S1)
Max. frame rate (FIFO<75%)	_____	LH5-A (S1)
Min. classifiable SNR (acc>80%)	_____	LH5-B (S3)
Max. classifiable distance (acc>80%)	_____	LH5-B (S3)
End-to-end latency at SF7 (mean±std)	_____	LH5-D (B1)
End-to-end latency at SF12 (mean±std)	_____	LH5-D (B3)
SF classification accuracy (nominal)	_____	LH5-D (B1)
Modulation classification accuracy (nominal)	_____	LH5-D (B1)
Emitter ID classification accuracy (nominal)	_____	LH5-D (B1)
Packet-type classification accuracy (nominal)	_____	LH5-D (B1)
Max. stable run time (tested)	_____	LH5-B (S4)
FPGA resource utilization (LUT/FF/BRAM)	_____	Table VIII
Host inference latency (p99)	_____	Table XV

ing any one of the seven is—per the LH5-D rule—"not a regression check; it is an exploration." The 5% threshold for declaring a regression at a fully pinned point is followed by a four-step investigation order (hardware first, firmware second, bitstream third, environment fourth) before the system itself is concluded to have degraded. Documenting the *absence* of a regression is itself a reportable result, because it provides empirical evidence that the platform is stable across module boundaries.

O. Module 5 System Specification Table

The headline Module 5 artifact is the system specification table populated entirely from measured experimental results, with every value cited back to the experiment that produced it. Table XIX lists the twelve specification rows the framework requires to be filled. Several rows are inherited from earlier modules—the FPGA resource utilization from Module 2 Table VIII, the host inference 99th-percentile from Module 4 Table XV—but the operating-envelope rows (maximum concurrent stations, maximum frame rate, minimum classifiable SNR, maximum classifiable distance, end-to-end latency at the SF7 and SF12 extremes, and maximum stable run time) are defined only after Module 5 measurements are available. This staging is what the framework means by "deriving the specification from evidence": the specification cannot be written until the stress and benchmark data exist, and conversely the stress and benchmark data cannot be assembled into a defensible specification without the comparability and uncertainty discipline imposed in LH5-C and LH5-D.

P. Module 5 PAUSE Behavioral Response and RSSI-Targeting Accuracy

The closed-loop intercept-and-act behavior of the PAUSE protocol is the framework's most visible cross-domain demonstration. Two structured experiments instrument it. The first measures the timing of the response loop: from the moment the operator presses the user button on the Threat board to the moment the targeted station prints `"I'm Jammed"` on its

TABLE XIX
MODULE 5 PAUSE PROTOCOL RESPONSE AND RECOVERY LATENCY ENVELOPE, WITH PER-COMPONENT CONTRIBUTIONS AND A BLANK "MEASURED" COLUMN FOR THE COHORT'S WALL-CLOCK LOG CAPTURE.

Quantity	Exp. (ms)	Meas. (ms)
PAUSE cadence (Threat TX interval)	100	—
PAUSE on-air time (SF7, 32 B)	∼50	_____
Target RX ISR + flag latency	<1	_____
Response latency (worst case)	∼151	_____
PAUSE hold-off duration (payload byte 1)	500	—
Target next-slot quantization (typical)	<1000	_____
Recovery latency (worst case)	∼1500	_____

TABLE XX
MODULE 5 RSSI-TARGETING ACCURACY EXPECTATION WITH TARGET A AT 1 M, TARGET B AT 5 M, THREAT AT A FIXED REFERENCE. THE "MEAS." COLUMN IS BLANK FOR THE COHORT'S TEN-TRIAL LOG.

Quantity	Exp.	Meas.
RSSI delta (Target A − Target B), dB	≥+10	_____
Trials targeting Target A out of 10	≥9	_____
`"I'm Jammed"` prints on Target A	≥9	_____
`"I'm Jammed"` prints on Target B	≤1	_____

USART2 debug log, and symmetrically from button release to the next successful DATA TX from the previously paused station. Both timestamps are captured to millisecond resolution against a common host wall clock so that the response and recovery latencies can be compared against the PAUSE TX cadence (100 ms) and the PAUSE hold-off (500 ms).

Table XX shows the expected response and recovery envelopes, decomposed by the dominant contribution. The response latency is bounded above by one full PAUSE cadence interval plus the SF7 airtime of the PAUSE packet itself plus the target's RX-side ISR processing; in the worst case (button pressed just after a PAUSE TX) the next PAUSE arrives 100 ms later, propagates over the air in ∼50 ms, and is processed in under 1 ms, for an upper bound of ∼151 ms. The recovery latency is bounded by the 500 ms hold-off from the last received PAUSE plus the target's next scheduled TX slot.

The second experiment measures RSSI-targeting accuracy with both Targets active at known distances (Target A at 1 m, Target B at 5 m) and the Threat at a fixed reference position. With ten button-press trials, the framework expects the Threat's per-Beacon-ID RSSI table to consistently rank Target A above Target B by at least 10 dB, and consequently for at least 9 of 10 trials to result in PAUSE packets addressed to Target A's ID and `"I'm Jammed"` prints appearing only on Target A's debug log. A failure of this expectation (e.g., the Threat picking Target B in trials where Target A's RSSI is briefly stale due to a missed packet) is itself diagnostically informative and traces to the staleness threshold in the Threat firmware (5-second default), which is a Module 1 design decision evaluated only at this Module 5 measurement point.

The two PAUSE experiments together exercise every cross-domain interface of Section II simultaneously: the PAUSE TX

TABLE XXI
MODULE 5 FAILURE-MODE CATALOG STARTER (LH5-E), WITH CHAIN DEPTH.

ID	Failure mode	Obs.	Root	Depth
F1	FIFO overflow (burst)	LH5-A	M2	1
F2	CRC reject on valid frame	LH3-C	M1/M2	1
F3	Low-SNR class. collapse	LH5-B	M3/M4	2
F4	Transient SF transition mis.	LH5-B	M2	1
F5	Label mismatch $>$ nominal	LH3-H	M1	2
F6	Op-condition shift accuracy drop	LH4-F	M1/M3	2
F7	Threat targets stale ID on PAUSE	LH5-pause	M1	2
F8	stdout buffering inflates lat.	LH5-C	M4	0

path uses the Threat firmware (firmware-to-RF), the targets receive over the same physical channel they normally transmit on (RF-to-firmware), the captured PAUSE frames flow through the FPGA and into the dataset (RF-to-FPGA, FPGA-to-data-pipeline), and the post-hoc analysis correlates dataset rows against the Threat's USART2 PAUSE event log to validate the closed-loop behavior end-to-end (data-to-model). No prior single-domain RF classification work supports this kind of measurement because none of them instrument an act stage in the loop; the PAUSE protocol is what makes the framework's intercept-and-act claim measurable rather than aspirational.

Q. Module 5 Failure-Mode Catalog and Chain-Depth Distribution

Lab Handout 5-E assembles the cross-domain failure-mode catalog that is the framework's most explicit pedagogical artifact. Each entry has five fields (ID, failure mode, observed-in handout, root module, root cause), with the root module being the *earliest* module whose design decision is responsible. A failure tagged with a root module different from its observed module is, by construction, a cross-domain integration issue—the very class of problem the framework exists to teach. Table XXII reproduces the eight starter entries the framework specifies as the minimum bar; the deliverable expands this list with locally observed failures, but never below eight.

Each entry is tagged with a chain depth: depth 0 if the root module equals the observed module (a local failure), depth 1 if the root is the immediate upstream neighbor (e.g., M1 causing an M2 symptom), and depth 2 or greater if the chain skips one or more module boundaries (e.g., an M1 firmware decision surfacing as an M4 accuracy problem or an M5 PAUSE-targeting error). The framework requires at least two depth-2 entries on the grounds that a catalog dominated by depth-0 entries indicates the curriculum's integration surface was too thin to develop the cross-domain reasoning skills that are the framework's primary objective. The starter table contains four depth-2 entries (F3, F5, F6, F7), which is the empirical floor against which a student cohort's catalog can be compared: an instructor evaluating whether a particular cohort engaged with the framework as intended can simply count depth-2 entries in their catalog, with the starter's four serving as the calibration baseline. A one-page depth-2 walkthrough of a single chosen entry is required as the deliverable's qualitative companion, articulating the full causal chain from the originating design decision to the observed symptom.

R. Module 5 Acceptance Criteria as Quantitative Outcomes

The Module 5 analyses in Sections IV-M–IV-R are tied together by a parallel set of pass/fail criteria, each mapped one-to-one onto a cross-domain interface from Section II. A student cohort completes Module 5 successfully when, across all lab groups: (i) the LH5-A stress test plan is filed with pre-declared pass/fail thresholds for all four Sx rows of Table XVI, the five S1 station-density configurations are run for the full five minutes each on the production bitstream with FIFO peak below 75% and PER below 5% at the declared maximum concurrent station count, and an ILA capture of the S1e burst is collected on the debug bitstream with the production bitstream restored before any subsequent measurement; (ii) the LH5-B S2 rapid-change test reproduces the predicted "$\leq$2 transient misclassifications per SF transition" bound traceable to the FPGA inter-arrival-time feature of Section III-B, the S3 sweep produces an accuracy-vs-SNR curve overlaying the LH4-G stratified-accuracy curve and labels the degradation mode as graceful or catastrophic, and the S4 endurance run completes 30 uninterrupted minutes with no FIFO overflow alarm and per-minute classification accuracy stable to within the LH4-G Wilson half-width; (iii) the LH5-C six-stage latency decomposition of (10) sums to T_{total} within propagated uncertainty (11), the LH5-D six-point benchmarking matrix of Table XVIII reports mean/std/p99 and dominant stage at each operating point with all seven comparability axes pinned for the B4 baseline-regression check, and the LH5-pause measurements of Tables XX and XXI are completed with response latency under 200 ms, recovery latency under 1500 ms, and at least 9 of 10 trials targeting the higher-RSSI station; and (iv) the LH5-D system specification table of Table XIX is populated entirely from measured data with each row cited to its source experiment, the LH5-E failure-mode catalog contains at least eight entries with at least two at depth 2 or greater, and the cross-domain dependency diagram and depth-2 walkthrough are included in the final deliverable. As with the Module 1, Module 3, and Module 4 criteria, a failure at any one of the four isolates the affected interface—empirical envelope, time-dependent stability, latency accountability, or cross-domain causation—for remediation without ambiguity.

V. COMPARISON TO SINGLE-DOMAIN PRIOR WORK

The works cited in Section III and reproduced in the bibliography below are the strongest available reference points for the technical building blocks this framework relies on, but they are not—and are not intended to be—curriculum frameworks. Each advances the state of the art within its own discipline by reporting peak accuracy, throughput, or resource utilization on a curated dataset under controlled conditions: Komkov et al. [1] report Boolean reservoir modulation classification competitive with CNNs at fewer trainable parameters; Maclellan et al. [3] report streaming CNN classification of eight modulation schemes on RFSoC at 128 MHz; Mutescu et al. [5] report

TABLE XXII
CATEGORICAL COMPARISON: FRAMEWORK OUTCOME METRICS VS. WHAT SINGLE-DOMAIN PRIOR WORK [1]–[8] REPORTS.

Metric	Framework	Prior work
Single-task peak accuracy on curated data	✓	✓
Multi-task accuracy with shared features	✓	—
Cross-module regression survival	✓	—
Failure-mode chain-depth distribution	✓	—
Closed-loop intercept-and-act latency	✓	—
Per-disciplinary-interface acceptance gating	✓	—

99.77% LoRa spreading-factor detection on spectrogram inputs across a -20 dB to +30 dB SNR range; Sagduyu and Erpek [6] report 93.3% LoRa device identification accuracy on raw I/Q under real experimental conditions; Dakic et al. [8] report neural-network LoRa demodulation matching coherent matched-filter performance with carrier-frequency-offset tolerance; Alnajjar et al. [4] report comparative DNN/CNN/DBN architecture studies for modulation classification; Lin et al. [2] report 28.2% angle-of-arrival inference cycle-count reduction with FPGA acceleration; and Dang et al. [7] report SoC FPGA LoRa gateway implementations within tight resource budgets. Each is excellent within its scope.

The framework described in this paper does not claim to outperform any of these works on the metrics they report. It is a different class of artifact: a curriculum that produces engineers capable of building, deploying, and reasoning about an integrated system, evaluated by metrics that the cited works architecturally cannot report because none of them is structured as a five-module dependency chain with cross-module measurement scaffolds. The four such metrics are summarized in Table XXIII and discussed below.

Multi-task accuracy with shared features. The framework trains four classifiers (SF, modulation, emitter ID, packet type) on disjoint forbidden-column subsets of the same 32-byte feature vector, with statistical significance computed via Wilson intervals (7) at the framework's expected $\sim$375-row test set per stratum. Each cited single-task work reports a single accuracy on a different dataset under different SNR distributions; cross-comparison of those numbers against the framework's per-task accuracy is not meaningful, but the framework's claim that all four tasks can be served from a single shared-feature pipeline operating end-to-end at sub-62 ms total response time (Table XV) is a system-level result none of the cited works attempts.

Cross-module regression survival. The LH5-D B4 baseline regression check (Section IV-O) requires the system to retain Module 3's PER and Module 4's accuracy after the Module 5 stress program, with all seven comparability axes pinned. A passing regression check is empirical evidence that the framework's cross-module interfaces (the radio packet, the forwarding frame, the feature vector) survived integration without silent drift. No cited work has this kind of result because none of them integrates four disciplines into a multi-module dependency chain.

Failure-mode chain-depth distribution. The LH5-E catalog (Section IV-R) is the most direct measurement of cross-domain learning the framework offers. The chain-depth histogram—specifically the count of depth $\geq$2 entries against the framework-imposed floor of two—measures whether students engaged with the integration surface or settled for module-local debugging. A cohort that produces eight entries all at depth 0 has, by construction, not learned to reason across domain boundaries; a cohort that produces multiple depth-2 chains has empirical evidence of cross-domain causal reasoning. Single-domain studies, by definition, cannot produce a multi-module failure chain because they have no upstream module to root-cause to.

Closed-loop intercept-and-act latency. The PAUSE behavioral measurements of Section IV-Q demonstrate a bounded response loop from intercepted RF observation, through interceptor decision (highest-RSSI target selection), back through an RF transmit action, to a measurable behavioral change at the targeted station. The cited classification works [1], [3]–[6], [8] all stop at the classification output, because that is the appropriate scope for their contribution; they do not close the loop back to RF action and do not report response latencies because they have no act stage. The framework's closed-loop measurement is what makes the platform an EW-style intercept-and-act demonstrator rather than a passive classifier.

Per-disciplinary-interface acceptance gating. The acceptance criteria of Sections IV-H, IV-L, and IV-S each map four pass/fail conditions one-to-one onto the four cross-domain interfaces of Section II. A failing module isolates the affected interface for remediation without ambiguity. This structure—deliberately built so that any single failed criterion identifies the responsible disciplinary boundary—is what makes the framework auditable as an instructional artifact. No single-domain work uses such a structure because it would have nothing to gate against; an accuracy number on a single curated dataset has no internal interfaces to map criteria to.

The honest summary of the comparison is therefore that the cited single-domain works are necessary technical references but insufficient as engineering education in the multidisciplinary contexts the framework targets. They report what their disciplines have learned to optimize; they do not develop, and cannot evaluate, the cross-domain reasoning that produces working systems. The framework is the answer to a question the cited works do not pose, and it is evaluated against metrics those works cannot report on. This is not a critique of their technical contribution—each is excellent within its scope—but it is a direct claim that single-domain accuracy numbers, no matter how high, do not constitute a curriculum, and that the metrics this framework instruments are categorically distinct from and complementary to what single-domain studies provide.

VI. CONCLUSION

This paper presented an educational framework that treats the interfaces between FPGA digital design, RF communications, embedded firmware, and machine learning as first-class

teaching objects. The framework is instantiated by a benchtop platform built from commercially available components—two STM32 Nucleo-L476RG cooperative target stations exchanging DATA and ACK traffic, one STM32 Nucleo-L476RG interceptor (Threat) station that passively captures the bidirectional traffic and operates a button-gated PAUSE protocol against the highest-RSSI target, RFM95W LoRa transceivers, and a Digilent Nexys A7-100T FPGA—and a five-module progressive curriculum that sequences students through hardware bring-up, RTL design, system integration, AI-driven classification, and full-system stress testing and benchmarking. Expected-results analysis of all five fully instantiated modules indicates that the chosen hardware should leave more than 100 dB of link margin at bench distances while still exposing every LoRa physical-layer parameter as a tunable student-facing variable, and that the complete Module 2 ingestion pipeline should close timing at the target 100 MHz with worst negative slack greater than 3 ns and use approximately 2% of available block RAM. For Module 3, the closed-form latency budget predicts that RF on-air time will account for $\sim$91% of end-to-end latency at SF7 and $\sim$99.7% at SF12, isolating the system's optimization headroom to the RF subsystem rather than the digital processing pipeline; the 268 fps UART throughput ceiling sits well above the operating envelope of three concurrent stations, and the 64-entry receive FIFO is expected to retain headroom even under maximum-contention ALOHA arrivals; the five sweep campaigns of Lab Handouts 3-F and 3-G target 4,900 transmitted packets across 21 collection runs, with an expected yield of at least 2,500 effective feature vectors with stratified train/val/test splits keyed on `run_id`; and every quantitative claim is paired with a measurement table whose actual-value column the cohort fills during execution. For Module 4, the dataset is gated by a ten-check audit that runs before any preprocessing, four classification tasks (SF detection, modulation, emitter ID, packet type) proceed on disjoint feature sets derived from a label-leakage table that a logistic-regression sanity check is required to validate at $\geq$0.95 accuracy on forbidden columns alone, four classifier families (DummyClassifier, decision tree, grid-searched random forest, fully-connected MLP, and 1-D CNN over 8-row sequences) are to be compared with 95% Wilson confidence intervals to make significance claims defensible at $n \approx 375$, the operating-condition shift gap $\Delta = \mathrm{Acc}(\mathrm{clean} \rightarrow \mathrm{clean}) - \mathrm{Acc}(\mathrm{clean} \rightarrow \mathrm{contended})$ is to be measured per task entirely on hardware data with the framework's expected outcome being recovery of at least half of Δ on the SF detection task after fine-tuning only the final linear layer on 10–20% of contended-regime data, and the multi-task real-time inference engine targets a 99th-percentile latency under 8 ms against a 996 ms inter-packet budget while a schema manifest enforces the LH4-A data contract at deployment time as well as at training time. For Module 5, the system specification is to be derived empirically from a four-dimensional stress matrix—station density, rapid SF change, low-SNR sweep, and 30-minute endurance—each row of which is bound to a pre-declared pass/fail threshold and probes a specific upstream design decision rather than running a generic "does it still work" check; the six-stage end-to-end latency decomposition $T_{\mathrm{total}} = T_{\mathrm{RF}} + T_{\mathrm{RX_FW}} + T_{\mathrm{UART_in}} + T_{\mathrm{FPGA}} + T_{\mathrm{UART_out}} + T_{\mathrm{infer}}$ is to be captured per operating point with explicit uncertainty propagation into the interceptor-firmware residual stage, and the six-point benchmarking matrix B1–B6 is expected to demonstrate RF-stage dominance at every operating point—supporting the architectural conclusion that on-FPGA inference does not reduce T_{total} unless the host is eliminated entirely; a baseline-regression check at the B4 operating point pins seven comparability axes (SF, station count and mode, distance, transmit power, firmware commit, bitstream hash, model artifact hash) before any cross-module comparison is admissible; the closed-loop PAUSE behavioral measurements bound the response latency under $\sim$151 ms and recovery latency under $\sim$1500 ms while requiring the Threat to correctly target the higher-RSSI station in at least 9 of 10 trials; and the LH5-E failure-mode catalog is required to contain at least eight cross-domain failures organized by chain depth, with at least two depth-2 entries used as the empirical evidence that the curriculum's integration surface developed the cross-domain reasoning the framework exists to teach. The Module 1 ingestion skeleton is deliberately minimal so that the CRC-16 checker, feature extraction datapath, and streaming inference blocks added in Modules 2 and 4 can be introduced as student additions rather than replacements—preserving the framework's integration-first pedagogy through the entire module sequence. Finally, Section V situates the framework's outcomes against the single-domain prior work cited throughout: the framework does not claim to outperform single-task peak accuracy results in the literature, and instead claims a categorically distinct contribution—multi-task accuracy on shared hardware features, cross-module regression survival, failure-mode chain-depth distribution, closed-loop intercept-and-act latency, and per-disciplinary-interface acceptance gating—that single-domain studies are architecturally unable to report. Taken together, these expected results indicate that the framework is pedagogically coherent and that, given the closed-form analyses, datasheet parameters, and framework-defined targets above, it is technically feasible within the resource and complexity budgets of a graduate-level capstone course; an instructor adopting the framework at another institution can reproduce both the hardware platform and the quantitative acceptance criteria directly from the specification given here, and the measurement scaffolds in this paper provide the structured locations into which a delivering cohort fills its actual results.

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